

Week 6 Report Group 9 for DS340W

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Abstract—Real time win probability prediction has become increasingly valuable in sports analytics, yet existing systems provide limited interactivity and fail to incorporate comprehensive contextual features that influence game outcomes. This project develops a machine learning based sports analytics dashboard that provides continuously updated win probabilities during live NBA games while introducing a novel interactive what if simulation capability. Our system integrates multiple data sources including play by play event data, team Elo ratings, dynamic player statistics, and real time lineup compositions to capture both pregame team strength and in game momentum shifts. We employ ensemble learning techniques, combining logistic regression, gradient boosting, and neural networks, to predict win probabilities that update in real time as game events unfold. The dashboard’s interactive simulation module enables users to explore scenarios such as alternative substitution patterns, hypothetical scoring sequences, or foul situations and immediately observe their projected impact on win probability. Additionally, we integrate large language models to generate natural language explanations of probability shifts, making complex analytics accessible to both casual fans and professional analysts.

Index Terms—Sports analytics, real-time prediction, dashboard visualization, win probability, interactive simulation.

I. INTRODUCTION

THE purpose of this project is to create a data science solution that enhances the way live sports are experienced by both fans and analysts. Current trackers provide raw statistics and static win probability curves, but lack interactivity and predictive depth. For instance, ESPN’s Basketball Power Index (BPI) [2] offers pregame win probabilities but does not update dynamically based on in-game events or allow users to explore hypothetical scenarios. Similarly, FiveThirtyEight’s NBA predictions [3] focus primarily on season-long forecasting and pregame probabilities rather than real-time, play-by-play probability updates. While Inpredictable [6] developed a functional in-game win probability calculator for the NBA, it remains a passive tool that displays probabilities without enabling user interaction or scenario exploration. These existing systems, though valuable for providing baseline win expectations, do not offer the participatory element of simulating alternative game trajectories or explaining probability shifts in accessible terms, leaving a significant gap in how fans and analysts can engage with predictive analytics during live games. Our dashboard will fill this gap by offering continuously updated winning probabilities, presented through intuitive visualizations.

Beneficiaries of this system include:

- Fans, who will gain engaging, real-time insight into how likely their team is to win.
- Sports media and commentators, who can enhance live broadcasts with predictive analytics.

- Teams and analysts, who can benchmark in-game decisions against probability forecasts.

The novel aspect of our work is the interactive “what-if” simulator, which enables users to adjust game factors and immediately see projected changes in win probability. This adds a participatory layer that does not exist in existing win probability dashboards.

In addition, we plan to leverage large language models (LLMs) to provide natural-language explanations of probability shifts and game scenarios, making analytics more accessible to fans and analysts.

A. Anticipated Technical Challenges

Several technical challenges are expected:

- 1) Real-time data ingestion from NBA APIs.
- 2) Synchronizing and cleaning play-by-play data streams.
- 3) Training predictive models that update probabilities in seconds.
- 4) Scaling the system to handle multiple games simultaneously.
- 5) Integrating the “what-if” module without disrupting live updates.

B. Data Science Framework

Our proposed solution will follow the data science pipeline:

- 1) Data acquisition - Collecting live data via NBA APIs
- 2) Data preprocessing - Cleaning and aligning game events by time
- 3) Modeling - Training machine learning models (regressions, gradient boosting, neural networks, etc) on historical data in order to predict various outcome probabilities
- 4) LLM - Alongside predictive modeling, we will incorporate large language models through prompt engineering to automatically generate contextual insights. These models will translate raw probabilities and what-if scenarios into natural-language summaries, enabling users to understand not just the numbers but the reasoning behind them.
- 5) Integration of our novel aspect - Create and embed the what-if simulator for user interaction
- 6) Visualization - Development of the dashboard to show live odds and scenario outcomes
- 7) Deployment - Host on an advanced cluster in order to handle the stream of live data

II. LITERATURE REVIEW

Our literature review consisted

A. General Machine Learning Framework for Sports Prediction

Bunker and Thabtah [7] establish a comprehensive framework for applying machine learning to sport result prediction, with a particular emphasis on Artificial Neural Networks and their effectiveness in handling the non-linear relationships present in sports data. Their analysis identifies three critical data categories for successful predictive models: historical match outcomes, player performance metrics, and opposition characteristics. The authors highlight specific challenges in sports prediction that distinguish it from conventional classification tasks, including high dimensional feature spaces, class imbalance in outcome distributions, and the inherent dependencies between sequential game events. Their proposed framework addresses model evaluation through cross-validation strategies that account for sequential ordering, preventing data leakage from future information into training sets, a relevant consideration for live sports prediction systems.

B. NBA Win Probability Implementation

Unpredictable [6] presents a practical NBA win probability calculator utilizing locally weighted logistic regression, a non-parametric algorithm that assigns greater weight to training examples similar to the current game state. The model employs four input variables: game time remaining, current point differential, possession status, and Vegas point spread. This feature selection demonstrates the tradeoff between bias and variance in sports prediction modeling, where a minimal feature set reduces model complexity and overfitting risk while capturing the most informative game state characteristics. The inclusion of Vegas point spread is particularly noteworthy as it incorporates the collective predictions of betting markets as a pregame team strength indicator. This approach aligns with the principle of using external signals to enhance model predictions, similar to how Elo ratings can be integrated as prior probabilities in sports prediction modeling.

C. Cross-Sport Win Probability Modeling

Robberechts [5] develops an in game win probability model for soccer using machine learning techniques that emphasize feature engineering and sequential modeling. His work addresses the challenge of capturing momentum and contextual game dynamics through time-series features and statistics that encode recent performance trends. The model uses gradient boosting methods, which are particularly effective for sports prediction due to their ability to model complex, nonlinear interactions between features while maintaining interpretability through feature importance metrics. This technique demonstrates the value of ensemble learning approaches that combine multiple weak learners to achieve robust predictions. Their framework also highlights the importance of probability calibration, ensuring that predicted probabilities accurately reflect true outcome frequencies.

Data storage and synchronization will be managed through structured databases to ensure scalability. Evaluation of early models will be benchmarked against naive baselines, such as assuming the team currently leading will win, to demonstrate clear improvement.

A. Week 6 Milestone

We have begun to set up a real-time data ingestion pipeline using the NBA API Python package, focusing initially on the NBA domain for proof of concept. The following endpoints will be integrated:

- **Play-by-play event logs:** Sequential time-stamped records of every action in a game.
- **Hustle stats:** Advanced metrics such as deflections, contested shots, and box-outs.
- **League standings and game context:** To add metadata to each simulation.

1) *Preprocessing:* Our preprocessing step focused on cleaning and aligning the play-by-play logs, which tend to contain inconsistencies and missing values due to the rapid logging format used by sports data feeds. This also included combining team and player stats to the play by play logs to allow for a more robust array of features in modeling. The data comes from multiple subpackages within the Python NBA API library. Datasets used include:

- **PlayByPlayV3** - A comprehensive dataset from the `nba_api` package containing detailed play-by-play logs for every game of the NBA season. This dataset serves as the foundation for predicting win probability in our models. Key features include unique game identifiers, time-stamped score updates, team information, and play descriptions that capture each significant event during the game.
- **ScoreboardV2** - Since many player and team statistical packages within the `nba_api` library require queries by either date or team identifier, data from this package was joined with the play-by-play dataset to enrich our feature set. This integration enables a seamless connection between game events and broader team and player statistics.
- **Elo Rating** - To quantify team strength based on historical performance and relative skill level, we incorporated each team's Elo rating as a predictive feature in our model. This metric provides an objective strength metric that biases predictions toward higher-rated teams, reflecting their competitive advantage. Our Elo data was collected from...
- **Teams** - A static reference table from the `nba_api` library containing essential team information, including full team names, standard abbreviations, and unique team identifiers. Since games in our Elo statistics dataset are only identified by team abbreviations and game date, this package was important in establishing proper joins across datasets and maintaining referential integrity across data sources.
- **LeagueDashPlayerStats** - A dataset, queried by date, from the `nba_api` library that provides cumulative season statistics for each player in the league up to the specified

III. IMPLEMENTATION PLAN

We will leverage Python libraries such as pandas for data pre-processing, and scikit-learn to develop baseline models.

date. These statistics enable our model to account for the skill composition of each team's active lineup on the court, allowing for dynamic assessment of team strength based on which players are currently in play.

- **BoxScoreTraditionalV2** - This package provides complete lineup configurations and traditional box score statistics for each game of the NBA season. We utilize this dataset to initialize starting lineups and track lineup changes throughout each game. Since play descriptions in the play-by-play dataset denote substitutions using only player surnames, this dataset is essential for accurately identifying players and updating active lineups in real time. This will allow us to retrieve the corresponding players' cumulative season statistics.

This step is critical for the success of the modeling and simulation modules, as time-aligned and semantically consistent input is required for effective prediction. The messiness of raw play-by-play data has proven to be more severe than expected. While NBA API is structured well, inconsistency in naming conventions, event overlaps, and logging granularity have made it difficult to cleanly segment actions. We have had to spend significant time designing preprocessing scripts that correct or realign these logs.

2) *Initial Models:*

- 1) Implement baseline logistic regression for win probability with a target of 60% accuracy
- 2) Develop a linear regression model for score prediction
- 3) Create model evaluation framework and performance tracking

Novel Aspect Integration

- 1) Create "what-if" simulator prototype allowing for parameter adjustments

IV. CONCLUSION

By the end of the project, we will deliver a scalable, real-time sports analytics dashboard with both predictive and interactive components. This system not only enhances fan engagement but also contributes a novel feature to sports data science: a dynamic simulator that merges live analytics with hypothetical exploration. Looking ahead, we envision our system not only serving as a fan engagement tool but also as a foundation for advanced decision support systems in sports analytics. By bridging live predictive modeling with interactive simulation, this project sets the stage for future innovations in real-time strategy optimization, broadcast enhancements, and even sports betting integrity analysis. The integration of LLMs ensures that our dashboard communicates results in a human-readable way, bridging the gap between complex predictive analytics and fan engagement. This capability positions the project as not only a quantitative tool but also a narrative platform for real-time sports analysis.

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